

review of the terminology concerning subatomic particles.

all particles are either...

	Bosons (integer spin)	Fermions (half-integer spin)	cf. spin-statistics theorem.
$m = 0$	photons, gluons ($s=1$) gravitons ($s=2$)	...	
$m \neq 0$	...	electron ($s = \frac{1}{2}$)	

[1] Folland, G.B. Quantum field theory: a tourist guide for mathematicians.

[2] Lee, John M. Introduction to Smooth Manifolds.

Single-particle space.

[3] relativistic invariance implies that the state space $\mathcal{H}$ must come equipped with

a projective unitary representation of the group $\mathbb{R}^{1,3} \ltimes \text{SO}^\uparrow(1,3)$.

Moreover, every projective unitary representation of $\mathbb{R}^{1,3} \ltimes \text{SO}^\uparrow(1,3)$ comes from
genuine unitary representation of its universal cover $\mathbb{R}^{1,3} \ltimes \text{SL}(2, \mathbb{C})$

(We shall assume that this representation is irreducible, which means
that the particle is in some sense "elementary".)

In [1], we can study the irreducible unitary representation of $\mathbb{R}^{1,3} \ltimes \text{SL}(2, \mathbb{C})$.

And that covers general cases ($s \geq 0, m \neq 0$). But I will follow our textbook.

For a model for electrons, we set

$$m = (\text{the rest mass of the electron}), \quad \mathcal{H} = L^2(X_m, \text{dim}, \mathbb{C}^2)$$

$$U: \rho \rightarrow \{U(A) : \forall A, \langle A \rangle \in \rho\}$$

$$\mathbb{R}^{1,3} \rtimes \text{SO}^+(1,3)$$

Q. Inner product? the repn.^U of the Poincaré group., ρ ?

To answer, we modify the (5) postulate in Lec. 11.

↓

"If an observer in a given coordinate system observes the system to be in state ψ , then an observer in a new coordinate system obtained by the action of (u, A) on $\mathbb{R}^{1,3}$ observes the state of the same system as $U(u, A)\psi$."

Here, U is a representation $\rightarrow U(a, A)(b, B) = U(a, A)U(b, B)$

Note expectations of observables under a state ψ are Invariant under multiplication of the state by a phase $\alpha \in \mathbb{C}$, ($|\alpha|=1$, (See Lec 3)

it doesn't violate the postulate (4).

?

Thus it suffices that U is unitary and a projective representation,

meaning that

$$U((u, A)(b, B)) = r(u, A, b, B) U(u, A) U(b, B), \quad r(u, A, b, B) \in \{z \in \mathbb{C} : |z|=1\}$$

Recall p.38 $\mathcal{P} = \mathbb{R}^{1,3} \times \mathcal{SO}^+(1,3)$.

The Lorentz group $O(1,3)$: the group of all linear transformation of $\mathbb{R}^{1,3}$ that preserve the Lorentz Inner product:

$$A \in O(1,3) \Leftrightarrow (Ax)_\mu (Ay)^\mu = \eta_{\mu\nu} y^\mu \text{ for all } x, y \Leftrightarrow A^\dagger g A = A$$

Conjugate transpose.

- $O(1,3)$ has 4 connected components, which are determined by values of two homomorphisms $O(1,3) \rightarrow \{\pm 1\}$

$$\begin{cases} A \mapsto \det A \\ A \mapsto \operatorname{sgn}(A^0_0) \end{cases}$$

$\ker[A \mapsto \det A] = \mathcal{SO}(1,3)$: the special Lorentz group

$\ker[A \mapsto \operatorname{sgn}(A^0_0)] = O^\uparrow(1,3)$: the orthochronous Lorentz group.

$\mathcal{SO}^\uparrow(1,3) = \mathcal{SO}(1,3) \cap O^\uparrow(1,3)$: the restricted Lorentz group

; the connected component of the identity of $O(1,3)$.

25.2 $SL(2, \mathbb{C})$: 2×2 complex matrices of determinant 1.

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad : \text{Pauli matrices.}$$

form a basis for the space $\mathcal{H}$ of 2×2 Hermitian matrices.

Consider a bijection $M: \mathbb{R}^{1,3} \rightarrow \mathcal{H}$

$$x \in \mathbb{R}^{1,3} \quad \mapsto \quad M(x) = \begin{pmatrix} x^0 + x^3 & x^1 - ix^2 \\ x^1 + ix^2 & x^0 - x^3 \end{pmatrix}$$

Note. $\det(M(x)) = x^2 \leftarrow \text{Minkowski}$

Every $A \in SL(2, \mathbb{C})$ acts on $\mathcal{H}$ by the map $X \mapsto AXA^\dagger$, we denote the corresponding action on $\mathbb{R}^{1,3}$ by $k(A)$:

$$M(k(A)x) = AM(x)A^\dagger$$

$$|A|=1$$

Note $\det(AHS) = \det(M_{k(A)}) = x^2$ & $k(A)$ linear for all $A \in SL(2, \mathbb{C})$.

Thus $k: SL(2, \mathbb{C}) \rightarrow O(1, 3)$. (norm preserving.)

$$\det(k(A)x) = \det(M(k(A)x)) = \det(AM(x)A^+) = x^2.$$

Facts.

* k is a 2-to-1 surjective homomorphism from $SL(2, \mathbb{C})$ onto $SO^+(1, 3)$.

(sketch) the differential of k at the identity, which takes $S \in \mathfrak{sl}(2, \mathbb{C})$ to

the map $M(x) \mapsto SM(x) + M(x)S^+$, is an isomorphism of Lie algebras ([1] p.11)

Since $SL(2, \mathbb{C})$ is connected, k factors through the connected component

$SO^+(1, 3)$ of $O(1, 3)$. Note that its kernel is $\{I, -I\}$. Thus k is

a Lie group homomorphism betw two connected Lie groups. Thus by

theorem ([2] theorem 2.31), k is a double covering of $SO^+(1, 3)$ by $SL(2, \mathbb{C})$.

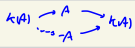
(This proof is based on [1].)

• " $k(A) = k(B)$ " iff " $A = B$ or $A = -B$ ". (We will write $A = \pm B$.)

* $k(A^+) = k(A)^+$ (omit the proof)

By *, $\exists \rho$ such that $k \circ \rho$ (as below) is the identity on $SO^+(1, 3)$.

$$SO^+(1, 3) \xrightarrow{\rho} SL(2, \mathbb{C}) \xrightarrow{k} SO^+(1, 3)$$

Note $\begin{cases} \rho(AB) = \pm \rho(A)\rho(B) \\ \rho(A^+) = \rho(A)^+ \end{cases}$ (**) 

25.3

Assume that there is a such p so that the projective repn. of $SO^\uparrow(1,3)$ in $\mathcal{H} = L^2(X_m, d\mu, \mathbb{C}^2)$ is given by

$$(U(a,A)\psi)(p) = e^{i\langle a,p \rangle} \rho(A) \psi(A^{-1}p), \quad (\text{fixed } (a,A) \in \mathcal{P})$$

• It is a projective repn.

$$\begin{aligned} U(a,A) \underbrace{(U(b,B)\psi)}_{\in \mathcal{H}}(p) &= e^{i\langle a,p \rangle} \rho(A) (U(b,B)\psi)(A^{-1}p) \quad \psi(\cdot) := \psi(A^{-1}\cdot) \\ &= e^{i\langle a,p \rangle} e^{i\langle b,p \rangle} \rho(A) \rho(B) \psi(B^{-1}A^{-1}p) \\ &= \pm e^{i\langle b-Ab,p \rangle} (U(a,A)U(b,B)\psi)(p) \end{aligned}$$

$$\begin{aligned} (a,A)(b,B) &= (a+Ab, AB) \\ &\quad (p.43) \end{aligned}$$

↑
Minkowski.

(1)

• To prove that $U(a,A)$ is unitary, we will define the inner product on $\mathcal{H}$.

Denote $p^* = (m, 0, 0, 0) \in X_m$. For any $p \in X_m$, $\exists$ maps taking p^* to p .

$\therefore$ The action of $SO^\uparrow(1,3)$ on X_m is transitive.

↑ In $X_m = \{p \in \mathbb{R}^{1,3} : p^* = m^2, p^0 > 0\}$, every element has the same mass, so that one can get a hyperbola btw any two points in X_m .

If A, B are any such maps, then

$$A = BR, \quad R = \begin{pmatrix} 1 & 0 \\ 0 & \mathcal{O} \end{pmatrix}, \quad \exists \mathcal{O} \in SO(3) \quad \text{Spatial rotation.}$$

$$(\because B \circ A^{-1}(p^*) = p^*; \text{ fix time. } \begin{bmatrix} 1 & 0 \\ 0 & \mathcal{O} \end{bmatrix} \begin{pmatrix} m \\ \vec{x} \end{pmatrix} = \begin{pmatrix} m \\ \vec{x} \end{pmatrix})$$

$\therefore \forall p \in X_m, \exists$ a unique positive-definite $V_p \in SL(2, \mathbb{C})$ such that $k(V_p)(p^*) = p$.

(Uniqueness is followed from the positive-definiteness)

$$\text{If } AA^\dagger = BB^\dagger \Rightarrow I = A^{-1}B(A^\dagger B^\dagger)^{-1}$$

p.113.

uniqueness

$$k(A') (p^*) = p \quad \text{as } A'$$

$$\rightarrow A' = AU', \quad U' \in SU(2)$$

o.b.z. $\frac{1}{2} \rightarrow B \frac{1}{2}$ fixed.

take $A \in SL_2(\mathbb{C})$ s.t. $k(A)(p^*) = p$.

$A = BU$: polar decomposition.

$SU(2)$ fixes p^* , we take $B = V_p$

$$k(BU)(p^*) = k(B)(p^*) = p.$$

$$SU(2) \hookrightarrow SL_2(\mathbb{C})$$

$$\begin{array}{ccc} k|_{SU(2)} & & \downarrow k \\ & & \end{array}$$

$$SO(3) \hookrightarrow SO^\uparrow(1,3)$$

$$A \in SL(2, \mathbb{C})$$

$$\rightarrow A = PU \rightarrow SU(2)$$

La positive definite Hermitian
($U^\dagger U = I \Rightarrow U \in SU(2)$)

Now we define the inner product:

$$(\psi, \varphi) = \int_{X_m} \psi(p)^{\dagger} V_p^{-2} \varphi(p) d\lambda_m(p), \quad \psi, \varphi \in \underbrace{L^2(X_m, d\lambda_m, \mathbb{C}^2)}_{\text{Hilbert space under this } (\cdot, \cdot)}$$

Claim: $U(a, A)$ is unitary; $(U(a, A)\psi, U(a, A)\varphi) = (\psi, \varphi)$

To this end, Note

$$(U(a, A)\psi)(p)^{\dagger} V_p^{-2} (U(a, A)\varphi)(p) = \psi(A^{-1}p)^{\dagger} \rho(A)^{\dagger} V_p^{-2} \rho(A) \varphi(A^{-1}p). \quad \text{by construction.}$$

$$(U(a, A)\psi, U(a, A)\varphi) = \int_{X_m} \psi(p)^{\dagger} \underbrace{\rho(A)^{\dagger} V_{A^{-1}p}^{-2} \rho(A)}_{\stackrel{?}{=} V_p^{-2}} \varphi(p) d\lambda_m(p). \quad \begin{aligned} \int_{X_m} \frac{d\lambda_m(p)}{2\sqrt{p}} f(p) &= \int_{\mathbb{R}^+} \frac{dt}{2\sqrt{t}} \frac{1}{2\sqrt{t}} f(A^{-1}p) \\ A &\in \text{SO}^+(1, 3) \end{aligned}$$

Lemma 25.1.

$$\rho(A)^{\dagger} V_{A^{-1}p}^{-2} \rho(A) = V_p^{-2}$$

Proof. $\kappa(V_{A^{-1}p}^{-2} \rho(A)) = \kappa(V_{A^{-1}p}^{-2}) \underbrace{\kappa(\rho(A))}_{=1} = \kappa(V_{A^{-1}p}^{-2}) A. \quad (\because \kappa: \text{homomorphism})$

$$A^{-1} \kappa(V_{A^{-1}p}^{-2}) (p^*) = A^{-1} A p = p \quad (\kappa(V_{A^{-1}p}^{-2})(p^*) = p)$$

$$\rightarrow A^{-1} \kappa(V_{A^{-1}p}^{-2}) = \kappa(V_p^{-2}) R \quad \text{for some spatial rotation } R.$$

$$\rightarrow \rho(A)^{-1} V_{A^{-1}p}^{-2} = \pm V_p^{-2} \rho(R). \quad (\times)$$

$$\therefore \underbrace{\rho(A)^{\dagger} V_{A^{-1}p}^{-2} \rho(A)}_{\text{positive-definite}} = \underbrace{V_p^{-1} (\rho(R)^{-1})^{\dagger}}_{\text{Inverse of } \rho(R)} \underbrace{\rho(R)^{-1} V_p^{-1}}_{\text{Inverse of } \rho(R)} = \pm V_p^{-2}$$

$\uparrow$
 $(\rho(R) \rho(R)^{\dagger})^{\dagger} = \pm (\rho(R)^{\dagger})^{\dagger} = \pm I.$

positive-definite
 $(x^{\dagger} \rho(A)^{\dagger} V_{A^{-1}p}^{-2} \rho(A) x = (\rho(A)x)^{\dagger} V_{A^{-1}p}^{-2} (\rho(A)x) > 0)$

$$\rho(A)^{\dagger} V_{A^{-1}p}^{-2} \rho(A) = V_p^{-2}.$$

///

25.4 Multi particle state space. (Dr. Shiu)

$\mathcal{H}$: the Hilbert space for a single fermionic particle

$$\mathcal{H}_{anti}^{\otimes n} = \left\{ \sum_{i_1, \dots, i_n \geq 1} \alpha_{i_1, \dots, i_n} e_{i_1} \otimes \dots \otimes e_{i_n} : \right. \\ \left. \forall \sigma \in S_n, \alpha_{i_{\sigma(1)}, \dots, i_{\sigma(n)}} = \text{sgn}(\sigma) \alpha_{i_1, \dots, i_n} \right\}$$

Alt

Note. $\alpha_{i_1, \dots, i_n} = 0$ if $\exists j \neq k, i_j = i_k$.

(Consider $(j, k) \in S_n$)

• basis-independent.

$$\mathcal{F}_0 = \bigoplus_{n=0}^{\infty} \mathcal{H}_{anti}^{\otimes n} \subseteq \mathcal{F}$$

dense

$$\mathcal{F}_1 = \left\{ (\psi_0, \psi_1, \dots) : \psi_n \in \mathcal{H}_{anti}^{\otimes n}, \sum_{n=0}^{\infty} \|\psi_n\|^2 < \infty \right\}$$

is the closure under the inner product

$$(\psi, \varphi) = \sum_{n=0}^{\infty} (\psi_n, \varphi_n) \quad \begin{cases} \psi = (\psi_0, \psi_1, \dots) \\ \varphi = (\varphi_0, \varphi_1, \dots) \end{cases}$$

We will take $\mathcal{H} = L^2(X_m, d\lambda_m, \mathbb{C}^2)$

Lec. 6.

for Bosonic particle

$$\mathcal{H}_{sym}^{\otimes n} = \left\{ \sum_{i_1, \dots, i_n \geq 1} \alpha_{i_1, \dots, i_n} e_{i_1} \otimes \dots \otimes e_{i_n} : \right. \\ \left. \forall \sigma \in S_n, \alpha_{i_{\sigma(1)}, \dots, i_{\sigma(n)}} = \alpha_{i_1, \dots, i_n} \right\}$$

$$\mathcal{B}_0 := \bigoplus_{n=0}^{\infty} \mathcal{H}_{sym}^{\otimes n} \subseteq \mathcal{B}$$

dense

$$\mathcal{B} = \left\{ (\psi_0, \psi_1, \dots) : \psi_n \in \mathcal{H}_{sym}^{\otimes n}, \sum_{n=0}^{\infty} \|\psi_n\|^2 < \infty \right\}$$

is the closure under the inner product

$$(\psi, \varphi) = \sum_{n=0}^{\infty} (\psi_n, \varphi_n).$$

26. The Dirac field.

26.1 A basis for fermionic Fock space

For an orthonormal basis $e_1, e_2, \dots$ of $\mathcal{H}$, can form an orthonormal basis for $\mathcal{H}_{\text{anti}}^{\otimes n}$ from elements of the form

$$\frac{1}{\sqrt{n!}} \sum_{\sigma \in S_n} \text{sgn}(\sigma) e_{i_{\sigma(1)}} \otimes \dots \otimes e_{i_{\sigma(n)}} = \sqrt{n!} e_{i_1} \wedge e_{i_2} \wedge \dots \wedge e_{i_n}.$$

where $i_1 < i_2 < \dots < i_n$. denote this by $|n_1, n_2, \dots\rangle$, where

$$n_j = \begin{cases} 1 & \text{if } j \in \{i_1, \dots, i_n\} \\ 0 & \text{o.w.} \end{cases}$$

ex) $n=3, i_1=1, i_2=2, i_3=3$

$$\frac{1}{\sqrt{3!}} \sum_{\sigma \in S_3} \text{sgn}(\sigma) e_{i_{\sigma(1)}} \otimes e_{i_{\sigma(2)}} \otimes e_{i_{\sigma(3)}}$$

$$= \frac{1}{\sqrt{3!}} (e_1 \otimes e_2 \otimes e_3 - e_2 \otimes e_1 \otimes e_3 - e_1 \otimes e_3 \otimes e_2 - e_3 \otimes e_2 \otimes e_1 + e_3 \otimes e_1 \otimes e_2 + e_2 \otimes e_3 \otimes e_1)$$

$$= |1, 1, 1, 0, 0, \dots\rangle$$

$|n_1, n_2, \dots\rangle$: n particles in states $e_{i_1}, \dots, e_{i_n}$.

* Pauli exclusion principle: the antisymmetry force to $n_j \in \{0, 1\}$: L2.

"No two identical fermions can occupy the same state."

* $\{|n_1, n_2, \dots\rangle : n_i \in \{0, 1\} \text{ \& \; } \sum n_i = n\}$: an orthonormal basis for $\mathcal{H}_{\text{anti}}^{\otimes n}$

$\{|n_1, n_2, \dots\rangle : n_i \in \{0, 1\} \text{ \& \; } \sum n_i < \infty\}$: an orthonormal basis for $\mathcal{F}$

26.2

$$k \geq 1. \quad \psi = |n_1, n_2, \dots\rangle$$

• the creation $a_k^\dagger$

$$a_k^\dagger \psi = \begin{cases} 0 & \text{if } n_k = 1, \\ (-1)^m |n_1, \dots, n_{k-1}, 1, n_{k+1}, \dots\rangle & \text{if } n_k = 0 \end{cases}$$

Zero of the vector space $\mathcal{H}$
(not the vacuum)

, $m = \#\{1 \leq i < k : n_i = 1\} = \sum_{i=1}^{k-1} n_i$

$$\otimes a_k^\dagger |0, 0, \dots\rangle = e_k$$

• the annihilation a_k ,

$$a_k \psi = \begin{cases} 0 & \text{if } n_k = 0 \\ (-1)^m |n_1, \dots, n_{k-1}, 0, n_{k+1}, \dots\rangle & \text{if } n_k \neq 0. \end{cases}$$

• Anticommutation relations: $\{A, B\} = AB + BA$

$$\{a_k, a_l^\dagger\} = \delta_{k,l} 1, \quad \{a_k^\dagger, a_l^\dagger\} = 0, \quad \{a_k, a_l\} = 0 \quad \text{for all } k, l \in \mathbb{N}.$$

i) $k=l$,

$$\{a_k, a_l^\dagger\} \psi = a_k a_l^\dagger \psi + a_l^\dagger a_k \psi = \begin{cases} \psi & n_k = 0 \\ \psi & n_k = 1 \end{cases}$$

$k \neq l$

$$\{a_k, a_l^\dagger\} \psi = \begin{cases} 0 \\ 0 \\ 0 \\ 0 \end{cases} \quad a_k \begin{bmatrix} n_k = 0, n_l = 0 \\ n_k = 0, n_l = 1 \\ n_k = 1, n_l = 0 \\ n_k = 1, n_l = 1 \end{bmatrix} a_l^\dagger$$

Continuous version of $a_k^\dagger, a_k$.

Define $A^\dagger, A: \mathcal{H} \rightarrow \mathcal{L}(\mathcal{H}_0, \mathcal{H})$: $f \in \mathcal{H}$, $f = \sum_{k=1}^{\infty} \alpha_k e_k$

$$A^\dagger(f)\psi := \sum_k \alpha_k a_k^\dagger \psi, \quad A(f)\psi := \sum_k \alpha_k^* a_k \psi.$$

* $A, A^\dagger$: basis independent

$$* \{A(\xi), A^\dagger(\eta)\} = (\xi, \eta) \mathbb{1}, \quad \{A^\dagger(\xi), A^\dagger(\eta)\} = 0, \quad \{A(\xi), A(\eta)\} = 0 \quad \text{for all } \xi, \eta \in \mathcal{H}.$$

Similar to p.31

Now if U is a unitary operator on $\mathcal{H}$, U induces a unitary operator on all the tensor powers on $\mathcal{H}$ and hence a unitary operator $\mathcal{F}(U)$ on the full tensor algebra $\mathcal{F}(\mathcal{H}) = \bigoplus_0^\infty \mathcal{H}^{\otimes k}$, by the formula

$$\mathcal{F}(U)(u_1 \otimes \dots \otimes u_k) = Uu_1 \otimes \dots \otimes Uu_k.$$

Using the commutativity of $\mathcal{F}(U)$ (with the number operator N and with the projections onto the symmetric and antisymmetric subspaces) and some auxiliary function,

We have

$$\mathcal{F}(U) \mathcal{F}(U)^\dagger = \mathbb{1} \quad \mathcal{F}(U)^\dagger \mathcal{F}(U) = \mathbb{1}$$

$$UA(\xi)U^\dagger = A(U\xi), \quad UA(\xi)^\dagger U^\dagger = A(U\xi)^\dagger \quad \text{for all } \xi \in \mathcal{H}.$$

$$(\mathcal{F}(U)^\dagger(u_1 \otimes \dots \otimes u_k) = U^\dagger u_1 \otimes \dots \otimes U^\dagger u_k)$$

See Lect 3 covered by Dr. Bae.

26.3 free field associated to electrons.

$\mathcal{H} = L^2(X_m, d\lambda_m, \mathbb{C}^2)$, $\mathcal{H}$: the fermion Fock space gr. to $\mathcal{H}$, e_1, e_2 : the standard basis of $\mathbb{C}^2$.

$p \in X_m$, $s \in \{1, 2\}$. For $\xi \in L^2(X_m, d\lambda_m)$

$$A^\dagger(\xi e_s) = \int_{X_m} d\lambda_m(p) \xi(p) a^\dagger(p, s), \quad A(\xi e_s) = \int_{X_m} d\lambda_m(p) \xi(p)^* a(p, s)$$

formally, $a^\dagger(p, s) = A^\dagger(\delta_p e_s)$, $a(p, s) = A(\delta_p e_s)$.

Recall that $X_m = \{p \in \mathbb{R}^3 \mid p^2 = m^2, p_0 \geq 0\} = \{(\omega_p, p) \mid p \in \mathbb{R}^3, \omega_p = \sqrt{m^2 + p^2}\}$

define

$$A^\dagger(p, s) := \frac{a^\dagger(p, s)}{\sqrt{2\omega_p}}, \quad a(p, s) = \frac{a(p, s)}{\sqrt{2\omega_p}},$$

so that

$$\{a(p, s), a^\dagger(q, t)\} = (2\pi)^3 \delta^{(3)}(p - q) \delta_{s,t} 1, \quad \{a(p, s), a(q, t)\} = 0 = \{a^\dagger(p, s), a^\dagger(q, t)\}$$

(:) $A^\dagger(\xi e_s) = \int_{X_m} d\lambda_m(p) \xi(p) \frac{1}{\sqrt{2\omega_p}} a^\dagger(p, s) = \int_{\mathbb{R}^3} \frac{d^3p}{(2\pi)^3} \cdot \frac{1}{\sqrt{2\omega_p}} \xi(\omega_p, p) a^\dagger(p, s)$ similar to p. 48

$$\iint \frac{d^3p d^3q}{(2\pi)^6} \frac{1}{\sqrt{2\omega_p}} \frac{1}{\sqrt{2\omega_q}} \xi(p) \eta(q)^* \{a(p, s), a^\dagger(q, t)\}$$

symbolic expression.
(p. 31-32)

$$= \{A(\xi e_s), A^\dagger(\eta e_s)\} = (\xi e_s, \eta e_s) 1 = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} \xi e_s(p) \eta^* e_s(p)$$

$$= \iint \frac{d^3p d^3q}{(2\pi)^6} \frac{1}{\sqrt{2\omega_p}} \frac{1}{\sqrt{2\omega_q}} (2\pi)^3 \delta^{(3)}(p - q) \delta_{s,t} \xi(p) \eta(q)^*$$

for all ξ, η .

Now recall that $p^* = (m, 0, 0, 0) \in X_m$, $k: SL(2, \mathbb{C}) \rightarrow SO^*(1, 3)$, and $\exists! V_p$, $k(V_p)p^* = p$.

Define $S_p \in GL(4, \mathbb{C})$ by

$$S_p := \begin{pmatrix} (V_p^+)^{-1} & 0 \\ 0 & V_p \end{pmatrix}.$$

Define a basis of $\mathbb{C}^4$:

$$f_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad f_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad f'_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad f'_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}.$$

Define

$$u(p, s) := S_p f_s = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix} \quad \text{and} \quad v(p, s) := S_p f'_s = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix}.$$

Let $\mathcal{G} = \mathcal{F} \otimes \mathcal{F}'$ where $\mathcal{F}, \mathcal{F}'$ are two copies of the Fermion Fock space for electrons, which denote the joint electron-position space.

↓
antiparticle counterpart of the electron. (same mass and spin but electric charge of $+1e$)

Let $b^\dagger, b$ on $\mathcal{F}' \sim a^\dagger, a$ on $\mathcal{F}$.

Note $\forall C: \mathcal{F} \otimes \mathcal{G}$ extends uniquely to $\mathcal{G}$ via $C(f \otimes f') = C f \otimes f'$.

For $k \in \{1, 2, 3, 4\}$, we define an operator-valued distribution ψ_k on $\mathbb{R}^{1,3}$ as

$$\psi_k(x) := \sum_{s=1}^2 \int_{\mathbb{R}^3} \frac{d^3 p}{(2\pi)^3} \frac{\sqrt{m}}{\sqrt{2\omega_p}} \left[e^{-i(x,p)} \underline{u_k(p,s)} \underline{a(p,s)} + e^{i(x,p)} \underline{v_k(p,s)} \underline{b^\dagger(p,s)} \right].$$

$\psi(x) := \begin{pmatrix} \psi_1(x) \\ \psi_2(x) \\ \psi_3(x) \\ \psi_4(x) \end{pmatrix}$: the (quantized) Dirac field. Compute ψ in p.51.
(Dirac spinor field) Further see [1] chp.5.